



LLM-Powered Data Cleaning Assistant

Executive Data Quality Report

Dataset	messy_customer.csv
Generated	01 October 2026
Time	10:38 PM
Report Type	Executive Summary

Executive Summary

AI summary unavailable

Data Quality Improvement

Metric	Value
Before Score	97.5
After Score	92.4
Improvement	-5.0999999999999994

Cleaning Metrics

Metric	Count
Emails Fixed	5
Countries Standardized	6
Duplicate Actions	1
Total Cleaning Actions	49

Cleaning Actions

#	Action
1	Row 1: Country: 'USA' → 'United States'
2	Row 1: Purchase_Amount: '\$120' → '120'
3	Row 2: Customer_Name: ' JOHN SMITH ' → 'John Smith'
4	Row 2: Age: '25 years' → '25'
5	Row 2: Gender: 'male' → 'Male'
6	Row 2: Country: 'USA' → 'United States'
7	Row 2: Purchase_Date: '10/01/2024' → '2024-01-10'
8	Row 3: Age: '-5' → '<NA>'
9	Row 3: Email: 'alicegmail.com' → '<NA>'
10	Row 3: Purchase_Amount: '■2,000' → '2000'
11	Row 3: Purchase_Date: '2024/02/15' → 'nan'
12	Row 4: Age: 'twenty' → '<NA>'
13	Row 4: Gender: 'F' → 'Female'
14	Row 4: Purchase_Date: '15-02-2024' → 'nan'
15	Row 5: Gender: 'MALE' → 'Male'

16	Row 5: Country: 'U.S.A' → 'United States'
17	Row 5: Purchase_Amount: '\$500' → '500'
18	Row 6: Customer_ID: '1005' → '1006'
19	Row 6: Customer_Name: 'Bob' → 'Charlie'
20	Row 6: Age: '35' → '42'
21	Row 6: Gender: 'M' → 'Female'
22	Row 6: Email: 'bob@yahoo.com' → '<NA>'
23	Row 6: Country: 'United States' → 'United Kingdom'
24	Row 6: Purchase_Amount: '500' → '800'
25	Row 6: Purchase_Date: '2024-03-01' → 'nan'
26	Row 7: Customer_ID: '1006' → '1007'
27	Row 7: Customer_Name: 'Charlie' → 'David'
28	Row 7: Age: '42' → '29'
29	Row 7: Gender: 'Female' → 'Male'
30	Row 7: Email: 'charlie@mail' → 'david@gmail.com'
31	Row 7: Country: 'UK' → 'United Kingdom'
32	Row 7: Purchase_Amount: '\$800' → '1000'
33	Row 7: Purchase_Date: 'March 5 2024' → '2024-03-08'
34	Row 8: Customer_ID: '1007' → '1008'
35	Row 8: Customer_Name: 'David' → 'Eva'
36	Row 8: Age: '29' → '<NA>'
37	Row 8: Gender: 'Male' → 'Female'
38	Row 8: Email: 'david@gmail.com' → 'eva@gmail.com'
39	Row 8: Country: 'United Kingdom' → 'India'
40	Row 8: Purchase_Amount: '1000' → '300'
41	Row 8: Purchase_Date: '2024-03-08' → 'nan'
42	Row 9: Customer_ID: '1008' → '1009'
43	Row 9: Customer_Name: 'Eva' → 'Mike'
44	Row 9: Age: 'nan' → '31'
45	Row 9: Gender: 'female' → 'Male'
46	Row 9: Email: 'eva@gmail.com' → 'mike@gmail.com'
47	Row 9: Purchase_Amount: '\$300' → '300'
48	Row 9: Purchase_Date: 'nan' → '2024-04-01'
49	Removed 1 duplicate customer(s).

